

Package ‘confeR’

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bayes_lin_reg_stats	<i>BCA linear regression summary stats</i>
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Description

Takes in dataframe with (X, y) and returns sufficient statistics ($X'X$, $X'y$, $y'y$, n)

Usage

```
bayes_lin_reg_stats(  
  df,  
  outcome,  
  covariates,  
  weights = NULL,  
  k = 0,  
  n_sites = 1  
)
```

Arguments

df	Dataframe containing covariates X and outcome y.
outcome	Name of outcome y.
covariates	Vector of covariate names.
weights	Vector of weights for weighted linear regression. Default NULL.
k	Site number in case local intercepts are used. Default 0 corresponds to not using local intercepts.
n_sites	Number of sites. Default 1.

Value

List with transmitted summary statistics for each local site.

Author(s)

Peter Degen

bca_glm_oneshot	<i>BCA GLM one-shot</i>
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Description

Perform one-shot computation of GLM parameter estimates given transmitted summary statistics. Users should use ‘bca_oneshot’, which wraps this method.

Usage

```
bca_glm_oneshot(sumstats, pooling = "none", glm_prior_lambda = 0, alpha = 0.05)
```

bca_iterate_sites	<i>BCA iterate over local sites</i>
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Description

Pre-processing wrapper for testing BCA methods using individual participant data.

Usage

```
bca_iterate_sites(
  model,
  family,
  data_split,
  use_local_intercepts,
  outcome = NULL,
  covariates = NULL,
  alpha = 0.05
)
```

Arguments

model	Formula object relating outcome to covariates.
family	Family to pass to glm() call, e.g. "gaussian".
data_split	List, each element containing a dataframe with the IPD from a local site.
use_local_intercepts	Logical. If true, use fixed site-specific intercepts for each local site.
outcome	Name of outcome y.
covariates	Vector of covariate names.
alpha	Numeric. Significance level for credible intervals. Default 0.05.
center_name	Character (optional). Name of covariate denoting site identity. Must be supplied if ‘use_local_intercepts’ is TRUE.

Value

List with transmitted summary statistics for each local site.

Author(s)

Peter Degen

bca_oneshot	<i>BCA one-shot</i>
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Description

Perform one-shot computation of parameter estimates given transmitted summary statistics.

Usage

```
bca_oneshot(  
  sumstats,  
  family,  
  use_local_intercepts,  
  use_local_variances = FALSE,  
  epsilon = 1e-10,  
  alpha = 0.05,  
  covariates = NULL,  
  glm_prior_lamda = 0  
)
```

Arguments

- sumstats List of summary statistics from each local site, e.g. obbject returned by ‘bca_iterate_sites’.
- family Family to pass to glm() call, e.g. "gaussian".
- use_local_intercepts Logical. If true, use fixed site-specific intercepts for each local site.
- use_local_variances Logical. If true, use fixed site-specific residual variances for each local site. Default FALSE.
- epsilon Numeric. Regularization for prior hyperparameters. Default: 1e-10.
- alpha Numeric. Significance level for credible intervals. Default: 0.05.
- glm_prior_lamda Numeric (optional). Prior for diagonal precision matrix in normal model, known variance case. See also Equation 7 in Jonker, Pazira and Coolen (2024). Default: 0.
- center_name Character (optional). Name of covariate denoting site identity. Must be supplied if ‘family’ is ‘gaussian’.

Value

List of oneshot parameter estimates

Author(s)

Peter Degen

forest_box

Multivariate forest plot wrapper

Description

Wrapper that computes pbbox and makes forest plot in one call.

Usage

```
forest_box(
  params_oneshot,
  sumstats,
  family,
  use_local_intercepts,
  pbboxes = NULL,
  center_name = "site",
  remove_intercept = FALSE,
  alpha = 0.05,
  remove_sigma2 = TRUE,
  ...
)
```

Arguments

`params_oneshot` Parameters returned by ‘bca_oneshot’.

`sumstats` List of summary statistics from each local site, e.g. object returned by ‘bca_iterate_sites’.

`family` Family to pass to `glm()` call, e.g. "gaussian".

`use_local_intercepts` Logical. If true, use fixed site-specific intercepts for each local site.

`center_name` Character. Name of covariate denoting site identity.

`alpha` Numeric. Significance level for credible intervals. Default 0.05.

`...` Further arguments passed to ‘prepare_forest_plot’.

Value

ggplot2 object.

Author(s)

Peter Degen

forest_plot

Multivariate forest plot

Description

Creates a forest plot with a facet for each covariate. Optionally prints Box's prior predictive tail probability for each site.

Usage

```
forest_plot(
  df_forest,
  pboxes = NULL,
  outfile = NULL,
  alpha = 0.05,
  nrow = 1,
  inline_plot = FALSE,
  use_log_scale = TRUE,
  order_box = TRUE
)
```

Arguments

df_forest	Dataframe returned by 'prepare_forest_plot'.
pboxes	Optional numeric vector of Box p-values, contained in list returned by 'box_check_all_sites'.
outfile	Optional character. If not null, save figure in this file.
alpha	Numeric. Significance level for credible intervals. Default 0.05.
nrow	Numeric. (default: 1)
inline_plot	Logical. If TRUE, adjust width and height of inline plot in Jupyter notebooks using 'options()'. (default: FALSE)
use_log_scale	Logical. If TRUE, print pbox as an S-value, i.e. $-\log_2(\text{pbox})$ (default: FALSE)
order_box	FALSE Logical. If TRUE, order sites by p-Box. (default: FALSE)

Value

ggplot2 object.

Author(s)

Peter Degen

get_pred_prob*Box prior predictive tail probability*

Description

Check if estimate from specific site is compatible with posterior from all other centers.

Usage

```
get_pred_prob(  
  center_identity,  
  sumstats,  
  n_sites,  
  reduced_params,  
  use_local_intercepts,  
  remove_intercept = FALSE  
)
```

Arguments

center_identity	The id of the site to be checked
sumstats	List of summary statistics from each local site, e.g. object returned by 'bca_iterate_sites'.
n_sites	Number of sites.
reduced_params	Reduced posterior parameters returned by 'get_reduced_params'.
use_local_intercepts	Logical. If true, use fixed site-specific intercepts for each local site.
remove_intercept	Logical. If true, ignore intercept in pbbox calculation. Default: FALSE.
covariates	Vector of covariate names.

Value

The Box prior predictive tail probability.

Author(s)

Peter Degen

get_reduced_params	<i>Get reduced params</i>
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Description

Use Reverse-Bayes to get reduced parameters obtained by leaving out one local site from the full posterior.

Usage

```
get_reduced_params(center_identity, params_oneshot, sumstats, family)
```

Arguments

center_identity	The id of the site to be removed.
params_oneshot	Parameters returned by bca_oneshot.
sumstats	List of summary statistics from each local site, e.g. obbject returned by 'bca_iterate_sites'.
family	Family to pass to glm() call, e.g. "gaussian".

Value

List with reduced parameter estimates.

Author(s)

Peter Degen

nurses_hom	<i>Nurses hom data</i>
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Description

This simulated data set consists of 1000 data points representing nurses from 25 hospitals. The outcome of interest is job-related stress among nurses (standardized) and the covariates are age (standardized), experience (standardized), sex (0 = male, 1 = female), the type of ward in which the nurse works (0 = general care, 1 = special care), and hospital (1, 2, ..., 25).

Usage

```
nurses_hom
```


Format

```
## 'nurses_hom'
```

data_ipd Individual participant data. A data frame with 1000 rows and 6 variables:

gender factor: 0 = male, 1 = female
age dbl: Standardized age
experience dbl: Standardized experience
wardtype factor: 0 = general care, 1 = special care
stress dbl: Standardized stress levels
hospital factor: Hospital ID, 1 to 25

summary_stats_hom List of summary statistics for each hospital: (X'X, X'y, y'y, n). Use this for models with a global intercept.

summary_stats_het List of summary statistics for each hospital: (X'X, X'y, y'y, n). Use this for models with a site-specific intercepts.

pred_check

Box prior predictive tail probabilities for all sites

Description

Calculate p_Box for all sites.

Usage

```
pred_check(
  params_oneshot,
  sumstats,
  family,
  use_local_intercepts,
  center_name = NULL,
  remove_intercept = FALSE
)
```

Arguments

params_oneshot Parameters returned by 'bca_oneshot'.

sumstats List of summary statistics from each local site, e.g. object returned by 'bca_iterate_sites'.

family Family to pass to glm() call, e.g. "gaussian".

use_local_intercepts Logical. If true, use fixed site-specific intercepts for each local site.

center_name Character. Name of covariate denoting site identity.

remove_intercept Logical. If true, ignore intercept in pbox calculation. Default: FALSE.

data_split The data split by local sites.

covariates Vector of covariate names.

n_sites Number of sites.

Value

List with pboxes and reduced parameters for all sites.

Author(s)

Peter Degen

prepare_forest_plot	<i>Prepare forest plot</i>
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Description

Get clean dataframe with local glm parameters from summary statistics for all sites plus the federated BCA estimates.

Usage

```
prepare_forest_plot(df_bca, sumstats, family, alpha = 0.05)
```

Arguments

df_bca	Dataframe with BCA estimates returned by ‘tidy_results’.
sumstats	List of summary statistics from each local site, e.g. object returned by ‘bca_iterate_sites’.
alpha	Numeric. Significance level for credible intervals. Default 0.05.

Value

Dataframe with parameter estimates and confidence intervals for all sites and covariates.

Author(s)

Peter Degen

tidy_results	<i>BCA tidy results</i>
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Description

Returns a tidy data frame of BCA results.

Usage

```
tidy_results(params_onehot, use_local_intercepts)
```

Arguments

- params_oneshot Parameters returned by ‘bca_oneshot’.
- use_local_intercepts Logical. If true, use fixed site-specific intercepts for each local site.

Value

Dataframe

Author(s)

Peter Degen

transform_X	<i>transform X</i>
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Description

Take data vector X and convert factors to numeric. Binary factors will be encoded by 0, 1. Factors with more than two levels will be transformed using one-hot encoding, dropping one level to avoid multicollinearity.

Usage

transform_X(X)

Arguments

- x Design matrix X with covariates

Value

Transformed design matrix X

Author(s)

Peter Degen

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